

ired": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "shape": "object")}]}

03 GET /parameters

RULES []

{ "request": { "body": null, "header": {}, "path": {}, "query": {} }, "responses": { "200": { "schema": { "additional_properties": false, "fields": { "LCL": { "nullable": true, "required": true, "schema": { "type": "number" } }, "LSL": { "nullable": true, "required": true, "schema": { "type": "number" } }, "TARGET": { "nullable": false, "required": true, "schema": { "type": "number" } }, "UCL": { "nullable": false, "required": true, "schema": { "type": "number" } }, "USL": { "nullable": false, "required": true, "schema": { "type": "number" } }, "area": { "nullable": false, "required": true, "schema": { "enum": ["Etch", "Photo"], "type": "string" }, "ctrl_lower": { "nullable": true, "required": true, "schema": { "type": "number" } }, "ctrl_upper": { "nullable": false, "required": true, "schema": { "type": "number" } }, "name": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "parameter_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "parameter_name": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "spec_lower": { "nullable": true, "required": true, "schema": { "type": "number" } }, "spec_upper": { "nullable": false, "required": true, "schema": { "type": "number" } }, "target_value": { "nullable": false, "required": true, "schema": { "type": "number" }, "unit": { "nullable": true, "required": true, "schema": { "type": "string" }, "upper_only": { "nullable": false, "required": true, "schema": { "type": "boolean" } }, "type": "object", "shape": "array", "503": { "schema": { "additional_properties": false, "fields": { "code": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "details": { "nullable": false, "required": true, "schema": { "additional_properties": true, "fields": {}, "type": "object" }, "message": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "type": "object", "shape": "object" } } }

04 POST /documents/search

RULES []

{ "request": { "body": { "additional_properties": false, "fields": { "model_code": { "nullable": true, "required": false, "schema": { "min_length": 1, "type": "string" }, "query": { "nullable": false, "required": true, "schema": { "max_length": 1000, "min_length": 1, "type": "string" }, "top_k": { "nullable": false, "required": false, "schema": { "default": 4, "maximum": 10, "minimum": 1, "type": "integer" } }, "type": "object", "header": {}, "path": {}, "query": {} }, "responses": { "200": { "schema": { "additional_properties": false, "fields": { "chunk_id": { "nullable": false, "required": true, "schema": { "pattern": "^+cs2[0-9]{4}\$", "type": "string" }, "content": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "doc_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "document_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "model_code": { "nullable": true, "required": true, "schema": { "min_length": 1, "type": "string" }, "score": { "nullable": false, "required": true, "schema": { "maximum": 1, "minimum": -1, "type": "number" }, "section": { "nullable": true, "required": true, "schema": { "type": "string" }, "title": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "type": "object", "shape": "array", "422": { "schema": { "additional_properties": false, "fields": { "code": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "details": { "nullable": false, "required": true, "schema": { "additional_properties": true, "fields": {}, "type": "object" }, "message": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "type": "object", "shape": "object", "503": { "schema": { "additional_properties": false, "fields": { "code": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "details": { "nullable": false, "required": true, "schema": { "additional_properties": true, "fields": {}, "type": "object" }, "message": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "type": "object", "shape": "object" } } }

05 GET /relations/chambers/{chamber_id}

RULES ["node_count equals len(nodes)", "relationship_count equals len(relationships)"]

{ "request": { "body": null, "header": {}, "path": { "chamber_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "query": { "label": { "nullable": false, "required": false, "schema": { "enum": ["Area", "Chamber", "Equipment", "EquipmentModel", "Parameter", "ProcessStep"], "type": "string" }, "limit": { "nullable": false, "required": false, "schema": { "default": 500, "maximum": 1000, "minimum": 1, "type": "integer" } }, "responses": { "200": { "schema": { "additional_properties": false, "fields": { "context": { "nullable": false, "required": true, "schema": { "additional_properties": false, "fields": { "adjacent_process_step_ids": { "nullable": false, "required": true, "schema": { "items": { "type": "string" }, "type": "array" }, "area": { "nullable": false, "required": true, "schema": { "enum": ["Etch", "Photo"], "type": "string" }, "chamber_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "equipment_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "model_code": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "parameter_ids": { "nullable": false, "required": true, "schema": { "items": { "type": "string" }, "type": "array" }, "process_step_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "relation_ids": { "nullable": false, "required": true, "schema": { "items": { "type": "string" }, "type": "array" } }, "type": "object", "graph_revision": { "nullable": false, "required": true, "schema": { "pattern": "^0-9a-fj{64}\$", "type": "string" }, "node_count": { "nullable": false, "required": true, "schema": { "minimum": 0, "type": "integer" }, "nodes": { "nullable": false, "required": true, "schema": { "items": { "additional_properties": false, "fields": { "business_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "label": { "nullable": false, "required": true, "schema": { "enum": ["Area", "Chamber", "Equipment", "EquipmentModel", "Parameter", "ProcessStep"], "type": "string" }, "name": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "node_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "properties": { "nullable": false, "required": true, "schema": { "additional_properties": true, "fields": {}, "type": "object" } }, "type": "object", "type": "array" }, "relationship_count": { "nullable": false, "required": true, "schema": { "minimum": 0, "type": "integer" }, "relationships": { "nullable": false, "required": true, "schema": { "items": { "additional_properties": false, "fields": { "from_node_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" }, "relation_id": { "nullable": false, "r

```
required": true, "schema": {"pattern": "^REL-[0-9a-f]{20}$", "type": "string"}}, "to_node_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object"}, "shape": "object"}, "404": {"schema": {"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "details": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "message": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "shape": "object"}, "422": {"schema": {"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "details": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "message": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "shape": "object"}, "503": {"schema": {"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "details": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "message": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "shape": "object"}}
```

06 GET /agent/runs

RULES []

```
{ "request": { "body": null, "header": {}, "path": {}, "query": { "date_from": {"nullable": false, "required": false, "schema": {"format": "date", "type": "string"}}, "date_to": {"nullable": false, "required": false, "schema": {"format": "date", "type": "string"}}, "predicted_fault_code": {"nullable": false, "required": false, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "status": {"nullable": false, "required": false, "schema": {"enum": ["COMPLETED", "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string"}}, "responses": { "200": { "schema": {"additional_properties": false, "fields": { "action_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "alarm_source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "approval_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "confidence": {"nullable": true, "required": true, "schema": {"maximum": 1, "minimum": 0, "type": "number"}}, "created_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "deliveries": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": { "channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, "type": "object"}, "type": "array"}}, "fault_code": {"nullable": true, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "fault_color": {"nullable": false, "required": true, "schema": {"type": "null"}}, "fault_name": {"nullable": false, "required": true, "schema": {"type": "null"}}, "latency_ms": {"nullable": false, "required": true, "schema": {"minimum": 0, "type": "integer"}}, "llm_model": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "predicted_fault_code": {"nullable": true, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "recommended_action": {"nullable": true, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["COMPLETED", "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string"}}, "tools": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": { "n": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result_summary": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "s": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, "tool_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object"}, "shape": "array"}, "422": {"schema": {"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "details": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "message": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "shape": "object"}, "503": {"schema": {"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "details": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "message": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "shape": "object"}}
```

07 POST /agent/ask

RULES [{"fault_code and ground_truth_fault_code are forbidden"}, {"metrology.alarm_result is forbidden"}]

```
{ "request": { "body": {"additional_properties": false, "fields": { "question": {"nullable": false, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "type": "object"}, "header": {}, "path": {}, "query": {}}, "responses": { "200": { "schema": {"additional_properties": false, "fields": { "answer": {"nullable": false, "required": true, "schema": {"type": "string"}}, "confidence": {"nullable": true, "required": true, "schema": {"maximum": 1, "minimum": 0, "type": "number"}}, "evidence": {"nullable": true, "required": true, "schema": {"additional_properties": false, "fields": { "chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "doc_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "section": {"nullable": true, "required": true, "schema": {"type": "string"}}, "type": "object"}}, "evidence_items": {"nullable": false, "required": true, "schema": {"items": {"discriminator": "type", "type": "discriminated_union", "variants": {"AL
```

[illegible]

```
uired": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "message": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object", "shape": "object"}}}
```

09 POST /approvals/{approval_id}/decision

RULES [APPROVED or REJECTED requires decided_by and decided_at, PENDING requires all decision fields null, only EQP_HOLD actions have approvals]

```
{ "request": { "body": { "additional_properties": false, "fields": { "decided_by": { "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "decision": { "nullable": false, "required": true, "schema": {"enum": [APPROVED, REJECTED], "type": "string"}}, "decision_comment": { "nullable": true, "required": false, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "type": "object", "header": {}, "path": { "approval_id": { "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "query": {}, "responses": { "200": { "schema": {"additional_properties": false, "fields": { "action_code": { "nullable": false, "required": true, "schema": {"enum": [EQP_HOLD, MONITORING, WARNING], "type": "string"}}, "action_id": { "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": { "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_id": { "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approved_at": { "nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "approved_by": { "nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber": { "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": { "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "created_at": { "nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_at": { "nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_by": { "nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "decision_comment": { "nullable": true, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "equipment": { "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": { "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "fault_code": { "nullable": false, "required": true, "schema": {"enum": [FOC, MFD, OTH, RFM, TMD], "type": "string"}}, "lot": { "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": { "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "predicted_fault_code": { "nullable": false, "required": true, "schema": {"enum": [FOC, MFD, OTH, RFM, TMD], "type": "string"}}, "reason": { "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "status": { "nullable": false, "required": true, "schema": {"enum": [APPROVED, PENDING, REJECTED], "type": "string"}}, "type": "object", "shape": "object", "404": { "schema": {"additional_properties": false, "fields": { "code": { "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "details": { "nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "message": { "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object", "shape": "object", "409": { "schema": {"additional_properties": false, "fields": { "code": { "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "details": { "nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "message": { "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object", "shape": "object", "422": { "schema": {"additional_properties": false, "fields": { "code": { "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "details": { "nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "message": { "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object", "shape": "object", "503": { "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "message": { "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object", "shape": "object" } }
```

10 GET /audit-logs

RULES []

```
{ "request": { "body": null, "header": {}, "path": {}, "query": { "actor_type": { "nullable": false, "required": false, "schema": {"format": "date", "type": "string"}}, "date_from": { "nullable": false, "required": false, "schema": {"format": "date", "type": "string"}}, "date_to": { "nullable": false, "required": false, "schema": {"format": "date", "type": "string"}}, "entity_id": { "nullable": false, "required": false, "schema": {"type": "string"}}, "entity_type": { "nullable": false, "required": false, "schema": {"type": "string"}}, "event_type": { "nullable": false, "required": false, "schema": {"type": "string"}}, "responses": { "200": { "schema": {"additional_properties": false, "fields": { "actor": { "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "actor_id": { "nullable": true, "required": true, "schema": {"type": "string"}}, "actor_type": { "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "after": { "nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "at": { "nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "audit_id": { "nullable": false, "required": true, "schema": {"minimum": 1, "type": "integer"}}, "before": { "nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "detail": { "nullable": true, "required": true, "schema": {"type": "string"}}, "entity": { "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "entity_id": { "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "entity_type": { "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "event": { "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "event_type": { "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "occurred_at": { "nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "type": "object", "shape": "array", "422": { "schema": {"additional_properties": false, "fields": { "code": { "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "details": { "nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "message": { "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object", "shape": "object" } }
```

```
e, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "shape": "object"}, "503": {"schema": {"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "details": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "message": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "shape": "object"}
}}
11 GET /dataset/bounds
RULES []
deferred inventory
12 GET /dashboard/summary
RULES []
deferred inventory
13 GET /alarms/{source}/{alarm_id}
RULES []
deferred inventory
14 GET /alarms/paged
RULES []
deferred inventory
15 GET /traces/catalog
RULES []
deferred inventory
16 POST /traces/search
RULES []
deferred inventory
17 GET /relations/equipment/{equipment_id}
RULES []
deferred inventory
18 GET /documents/{document_id}
RULES []
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19 POST /agent/runs
RULES []
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[illegible]

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21 POST /agent/runs/{run_id}/retry
RULES []
deferred inventory
22 GET /agent/runs/paged
RULES []
deferred inventory
23 GET /approvals/paged
RULES []
deferred inventory
24 GET /actions
RULES []
{"request": {"body": null, "header": {}, "path": {}, "query": {"action_code": {"nullable": false, "required": false, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, {"responses": {"200": {"schema": {"items": {"additional_properties": false, "fields": {"action_code": {"nullable": false, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, {"action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"approval_status": {"nullable": true, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, {"chamber": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"created_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, {"created_by_agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"deliveries": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES"], "type": "string"}}, {"status": {"nullable": false, "required": true, "schema": {"enum": ["BLOCKED", "CANCELLED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, {"type": "object"}}, {"type": "array"}}, {"equipment": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"equipment_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"lot": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"reason": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"type": "object"}, {"type": "array"}, {"shape": "array"}, {"422": {"schema": {"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"details": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {"type": "object"}}, {"message": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"type": "object"}, {"shape": "object"}, {"503": {"schema": {"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"details": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {"type": "object"}}, {"message": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"type": "object"}, {"shape": "object"}}}}
25 GET /actions/{action_id}
RULES []
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26 POST /actions/{action_id}/deliveries/{channel}/retry

RULES []

deferred inventory

27 POST /internal/actions/{action_id}/delivery

RULES [{"FAILED requires error_code non-null and permits provider_message_id null", "SENT requires provide
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28 POST /analytics/query

RULES []

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29 POST /analytics/graph-query

RULES []

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30 POST /analytics/validate

RULES []

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```
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31 GET /analytics/history

RULES []

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32 GET /analytics/evaluations

RULES []

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              "type": "string"
            },
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              "required": true,
              "schema": {
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                "fields": {
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                }
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                "required": true,
                "schema": {
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                  "type": "string"
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}

GET /audit-logs/paged
RULES []

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GET /health
RULES []

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GET /health/ready
RULES [{"all checks PASS yields status READY and HTTP 200", "any check FAIL yields status NOT_READY and HTTP 503"}]

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DTO COMPONENTS
ActionCode
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ActionItem

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AgentAskCompatibilityEvidence

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AgentAskRequest

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AgentAskResponse

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AgentRunAcceptedResponse

```
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AgentRunActionItem

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AgentRunApprovalItem

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AgentRunCreateRequest

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```

AgentRunDetailResponse

```
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AgentRunItem

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AlarmAskEvidence

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AlarmEvidence

```
{ "additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["ALARM"], "type": "string"}}, "type": "object"}
```

AlarmItem

```
{ "additional_properties": false, "fields": {"action_code": {"nullable": true, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "alarm_type": {"nullable": false, "required": true, "schema": {"enum": ["OOC", "OOS"], "type": "string"}}, "area": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chamber": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "cl": {"nullable": true, "required": true, "schema": {"type": "number"}}, "equipment": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "fault": {"nullable": true, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "lcl": {"nullable": true, "required": true, "schema": {"type": "number"}}, "lot": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "mes": {"nullable": false, "required": true, "schema": {"type": "string"}}, "mes_status": {"nullable": true, "required": true, "schema": {"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, "notify": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "notify_status": {"nullable": true, "required": true, "schema": {"enum": ["FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, "occurred_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "parameter": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "parameter_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "predicted_fault_code": {"nullable": true, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "recipe": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "recipe_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "recipe_step_no": {"nullable": false, "required": true, "schema": {"minimum": 1, "type": "integer"}}, "rule_code": {"nullable": false, "required": true, "schema": {"enum": ["R03_CONSEC", "SUMMARY_OOC", "TRACE_OOS"], "type": "string"}}, "seq_no": {"nullable": true, "required": true, "schema": {"minimum": 0, "type": "integer"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "statistic_type": {"nullable": true, "required": true, "schema": {"type": "string"}}, "step_no": {"nullable": false, "required": true, "schema": {"minimum": 1, "type": "integer"}}, "ucl": {"nullable": true, "required": true, "schema": {"type": "number"}}, "value": {"nullable": true, "required": true, "schema": {"type": "number"}}, "wafer": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "wafer_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "rules": ["source TRACE requires rule_code TRACE_OOS", "source SUMMARY requires rule_code SUMMARY_OOC", "source R03 requires alarm_type OOC, rule_code R03_CONSEC, and value null", "compatibility aliases derive only from canonical fields"], "type": "object"}
```

AlarmRef

```
{ "additional_properties": false, "fields": {"alarm_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}}, "type": "object"}
```

AlarmSource

```
{ "enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}
```

AnalysisQueryRequest

```
{
  "additional_properties": {
    "fields": {
      "question": {
        "nullable": false,
        "required": true,
        "schema": {
          "max_length": 1000,
          "min_length": 1,
          "type": "string"
        }
      }
    }
  },
  "type": "object"
}
```

AnalysisQueryResponse

```
{
  "additional_properties": {
    "fields": {
      "columns": {
        "nullable": false,
        "required": true,
        "schema": {
          "items": {
            "type": "string",
            "type": "array"
          },
          "cross_check": {
            "nullable": true,
            "required": false,
            "schema": {
              "additional_properties": {
                "fields": {
                  "cypher": {
                    "nullable": true,
                    "required": false,
                    "schema": {
                      "type": "string"
                    }
                  },
                  "status": {
                    "nullable": false,
                    "required": true,
                    "schema": {
                      "enum": [
                        "MATCH",
                        "MISMATCH",
                        "SKIPPED"
                      ],
                      "type": "string"
                    }
                  },
                  "summary": {
                    "nullable": true,
                    "required": false,
                    "schema": {
                      "type": "string"
                    }
                  },
                  "type": {
                    "type": "object",
                    "error_msg": {
                      "nullable": true,
                      "required": false,
                      "schema": {
                        "type": "string",
                        "generated_sql": {
                          "nullable": true,
                          "required": false,
                          "schema": {
                            "type": "string"
                          }
                        },
                        "group_by": {
                          "nullable": false,
                          "required": true,
                          "schema": {
                            "items": {
                              "type": "string",
                              "type": "array"
                            },
                            "is_rejected": {
                              "nullable": false,
                              "required": true,
                              "schema": {
                                "type": "boolean"
                              }
                            },
                            "is_valid": {
                              "nullable": false,
                              "required": true,
                              "schema": {
                                "type": "boolean"
                              }
                            },
                            "latency_ms": {
                              "nullable": false,
                              "required": true,
                              "schema": {
                                "minimum": 0.0,
                                "type": "integer"
                              }
                            },
                            "metric": {
                              "nullable": true,
                              "required": false,
                              "schema": {
                                "additional_properties": {
                                  "fields": {
                                    "column": {
                                      "nullable": true,
                                      "required": false,
                                      "schema": {
                                        "type": "string",
                                        "p": {
                                          "nullable": true,
                                          "required": false,
                                          "schema": {
                                            "maximum": 100.0,
                                            "minimum": 0.0,
                                            "type": "number"
                                          }
                                        },
                                        "type": {
                                          "nullable": false,
                                          "required": true,
                                          "schema": {
                                            "enum": [
                                              "count",
                                              "max",
                                              "mean",
                                              "median",
                                              "min",
                                              "percentile",
                                              "ratio",
                                              "std",
                                              "sum"
                                            ],
                                            "type": "string"
                                          }
                                        }
                                      },
                                      "type": "object"
                                    },
                                    "metric_result": {
                                      "nullable": false,
                                      "required": false,
                                      "schema": {
                                        "type": "union",
                                        "variants": [
                                          {
                                            "type": "integer"
                                          },
                                          {
                                            "type": "number"
                                          },
                                          {
                                            "items": {
                                              "additional_properties": {
                                                "fields": {
                                                  "group": {
                                                    "nullable": false,
                                                    "required": true,
                                                    "schema": {
                                                      "additional_properties": {
                                                        "fields": {
                                                          "type": "object"
                                                        },
                                                          "value": {
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                                                            "required": false,
                                                            "schema": {
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                                                                  "type": "integer"
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                                                                  "type": "null"
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                                                              ]
                                                            }
                                                          },
                                                          "type": "object"
                                                        },
                                                          "type": "array"
                                                        },
                                                          "type": "null"
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                                                  },
                                                  "nl_query_log_id": {
                                                    "nullable": true,
                                                    "required": false,
                                                    "schema": {
                                                      "minimum": 1.0,
                                                      "type": "integer"
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                                                  "question": {
                                                    "nullable": true,
                                                    "required": true,
                                                    "schema": {
                                                      "type": "string",
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                                                        "nullable": true,
                                                        "required": false,
                                                        "schema": {
                                                          "type": "string"
                                                        }
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                                                      "row_count": {
                                                        "nullable": false,
                                                        "required": true,
                                                        "schema": {
                                                          "minimum": 0.0,
                                                          "type": "integer"
                                                        }
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                                                      "rows": {
                                                        "nullable": false,
                                                        "required": true,
                                                        "schema": {
                                                          "items": {
                                                            "additional_properties": {
                                                              "fields": {
                                                                "type": "object"
                                                              },
                                                              "type": "array"
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                                                            "visualization": {
                                                              "nullable": true,
                                                              "required": false,
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                                                                "additional_properties": {
                                                                  "fields": {
                                                                    "chart_type": {
                                                                      "nullable": false,
                                                                      "required": true,
                                                                      "schema": {
                                                                        "enum": [
                                                                          "bar",
                                                                          "histogram",
                                                                          "line",
                                                                          "table"
                                                                        ],
                                                                        "type": "string"
                                                                      },
                                                                      "x": {
                                                                        "nullable": true,
                                                                        "required": false,
                                                                        "schema": {
                                                                          "type": "string"
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                                                                        "nullable": true,
                                                                        "required": false,
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                                                                          "type": "string"
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                                                                  "type": "object"
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                                                              "type": "object"
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                                                        }
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                                                  }
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                                        ]
                                      }
                                    }
                                  }
                                }
                              }
                            }
                          }
                        }
                      }
                    }
                  }
                }
              }
            }
          }
        }
      }
    }
  },
  "type": "object"
}
```

ApprovalDecisionRequest

```
{
  "additional_properties": {
    "fields": {
      "decided_by": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        }
      },
      "decision": {
        "nullable": false,
        "required": true,
        "schema": {
          "enum": [
            "APPROVED",
            "REJECTED"
          ],
          "type": "string"
        }
      },
      "decision_comment": {
        "nullable": true,
        "required": false,
        "schema": {
          "max_length": 1000,
          "min_length": 1,
          "type": "string"
        }
      }
    }
  },
  "type": "object"
}
```

ApprovalItem

```
{
  "additional_properties": {
    "fields": {
      "action_code": {
        "nullable": false,
        "required": true,
        "schema": {
          "enum": [
            "EQP_HOLD",
            "MONITORING",
            "WARNING"
          ],
          "type": "string"
        }
      },
      "action_id": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        }
      },
      "agent_run_id": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        }
      },
      "approval_id": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        }
      },
      "approved_at": {
        "nullable": true,
        "required": true,
        "schema": {
          "format": "date-time",
          "type": "string"
        }
      },
      "approved_by": {
        "nullable": true,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        }
      },
      "chamber": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        }
      },
      "chamber_id": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        }
      },
      "created_at": {
        "nullable": false,
        "required": true,
        "schema": {
          "format": "date-time",
          "type": "string"
        }
      },
      "decided_at": {
        "nullable": true,
        "required": true,
        "schema": {
          "format": "date-time",
          "type": "string"
        }
      },
      "decided_by": {
        "nullable": true,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        }
      },
      "decision_comment": {
        "nullable": true,
        "required": true,
        "schema": {
          "max_length": 1000,
          "min_length": 1,
          "type": "string"
        }
      },
      "equipment": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        }
      },
      "equipment_id": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        }
      },
      "fault_code": {
        "nullable": false,
        "required": true,
        "schema": {
          "enum": [
            "FOC",
            "MFD",
            "OTH",
            "RFM",
            "TMD"
          ],
          "type": "string"
        }
      },
      "lot": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        }
      },
      "lot_id": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        }
      },
      "predicted_fault_code": {
        "nullable": false,
        "required": true,
        "schema": {
          "enum": [
            "FOC",
            "MFD",
            "OTH",
            "RFM",
            "TMD"
          ],
          "type": "string"
        }
      },
      "reason": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        }
      },
      "status": {
        "nullable": false,
        "required": true,
        "schema": {
          "enum": [
            "APPROVED",
            "PENDING",
            "REJECTED"
          ],
          "type": "string"
        }
      }
    }
  },
  "rules": [
    "PENDING requires all decision fields null",
    "APPROVED or REJECTED requires decided_by and decided_at",
    "only EQP_HOLD actions have approvals"
  ],
  "type": "object"
}
```

AskDocumentEvidenceAlias

```
{
  "additional_properties": {
    "fields": {
      "chunk_id": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        }
      },
      "doc_id": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        }
      },
      "document_id": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        }
      },
      "section": {
        "nullable": true,
        "required": true,
        "schema": {
          "type": "string"
        }
      }
    }
  },
  "type": "object"
}
```

AskToolItem

```
{
  "additional_properties": {
    "fields": {
      "name": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        }
      },
      "result": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        }
      },
      "result_summary": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        }
      },
      "status": {
        "nullable": false,
        "required": true,
        "schema": {
          "enum": [
            "ERROR",
            "SUCCESS"
          ],
          "type": "string"
        }
      }
    }
  },
  "type": "object"
}
```

```
ESS", "TIMEOUT"], "type": "string"}}, "tool_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}
```

AuditLogItem

```
{"additional_properties": false, "fields": {"actor": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "actor_id": {"nullable": true, "required": true, "schema": {"type": "string"}}, "actor_type": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "after": {"nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {"type": "object"}}, "at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "audit_id": {"nullable": false, "required": true, "schema": {"minimum": 1, "type": "integer"}}, "before": {"nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {"type": "object"}}, "detail": {"nullable": true, "required": true, "schema": {"type": "string"}}, "entity": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "entity_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "entity_type": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "event": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "event_type": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "occurred_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "type": "object"}
```

AuditLogPageResponse

```
{"additional_properties": true, "fields": {"event_type_counts": {"nullable": false, "required": true, "schema": {"additional_properties": {"type": "integer"}, "fields": {"type": "object"}}, "event_types": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "items": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": true, "fields": {"actor": {"nullable": false, "required": true, "schema": {"type": "string"}}, "actor_id": {"nullable": true, "required": true, "schema": {"type": "string"}}, "actor_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "after": {"nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {"type": "object"}}, "at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "audit_id": {"nullable": false, "required": true, "schema": {"type": "integer"}}, "before": {"nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {"type": "object"}}, "detail": {"nullable": true, "required": true, "schema": {"type": "string"}}, "entity": {"nullable": false, "required": true, "schema": {"type": "string"}}, "entity_id": {"nullable": false, "required": true, "schema": {"type": "string"}}, "entity_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "event": {"nullable": false, "required": true, "schema": {"type": "string"}}, "event_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "occurred_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "type": "object"}}, "page": {"nullable": false, "required": true, "schema": {"type": "integer"}}, "size": {"nullable": false, "required": true, "schema": {"type": "integer"}}, "total": {"nullable": false, "required": true, "schema": {"type": "integer"}}, "type": "object"}
```

AutoToolCallItem

```
{"additional_properties": false, "fields": {"n": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result_summary": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "s": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, "tool_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}
```

ChamberGraphResponse

```
{"additional_properties": false, "fields": {"context": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"adjacent_process_step_ids": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "area": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "model_code": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "parameter_ids": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "process_step_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "relation_ids": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "type": "object"}}, "graph_revision": {"nullable": false, "required": true, "schema": {"pattern": "^[0-9a-f]{64}$", "type": "string"}}, "node_count": {"nullable": false, "required": true, "schema": {"minimum": 0, "type": "integer"}}, "nodes": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"business_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "label": {"nullable": false, "required": true, "schema": {"enum": ["Area", "Chamber", "Equipment", "EquipmentModel", "Parameter", "ProcessStep"], "type": "string"}}, "name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "node_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "properties": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {"type": "object"}}, "type": "object"}}, "relationship_count": {"nullable": false, "required": true, "schema": {"minimum": 0, "type": "integer"}}, "relationships": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"from_node_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "relation_id": {"nullable": false, "required": true, "schema": {"pattern": "^REL-[0-9a-f]{20}$", "type": "string"}}, "to_node_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}}, "type": "array"}}, "rules": ["node_count equals len(nodes)", "relationship_count equals len(relationships)], "type": "object"}
```

ChamberRelationResponse

```
{"additional_properties": false, "fields": {"graph_revision": {"nullable": false, "required": true, "sche
```

```
ma": {"min_length": 1, "type": "string"}}, "nodes": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"business_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "display_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "label": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "properties": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "type": "object"}, "type": "array"}, "relationships": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "target": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}, "root_node_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}
}
ChartType
{"enum": ["bar", "histogram", "line", "table"], "type": "string"}
ChatToolCallItem
{"additional_properties": false, "fields": {"name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result_summary": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, "tool_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}
CrossCheck
{"additional_properties": false, "fields": {"cypher": {"nullable": true, "required": false, "schema": {"type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["MATCH", "MISMATCH", "SKIPPED"], "type": "string"}}, "summary": {"nullable": true, "required": false, "schema": {"type": "string"}}, "type": "object"}
DeliveryCallbackRequest
{"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES MOCK"], "type": "string"}}, "completed_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "error_code": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "event_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "provider_message_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "request_hash": {"nullable": false, "required": true, "schema": {"max_length": 64, "min_length": 64, "pattern": "^[0-9a-f]{64}$", "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["FAILED", "SENT"], "type": "string"}}, "rules": ["SENT requires provider_message_id non-null and error_code null", "FAILED requires error_code non-null and permits provider_message_id null"], "type": "object"}
DeliveryChannel
{"enum": ["EMAIL", "MES MOCK"], "type": "string"}
DeliveryResult
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DeliveryStatus
{"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}
DocumentAskEvidence
{"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "section": {"nullable": true, "required": true, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}
DocumentChunkItem
{"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chunk_seq": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "content": {"nullable": false, "required": true, "schema": {"type": "string"}}, "section_title": {"nullable": true, "required": false, "schema": {"type": "string"}}, "type": "object"}
DocumentDetailResponse
{"additional_properties": false, "fields": {"chunks": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chunk_seq": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "content": {"nullable": false, "required": true, "schema": {"type":
```

```
"string"}}, "section_title": {"nullable": true, "required": false, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "doc_type": {"nullable": true, "required": false, "schema": {"enum": ["MANUAL", "SPEC", "TROUBLESHOOT"], "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "model_code": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "source_path": {"nullable": true, "required": false, "schema": {"type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"type": "string"}}, "version": {"nullable": true, "required": false, "schema": {"type": "string"}}, "type": "object"}
```

DocumentEvidence

```
{"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "excerpt": {"nullable": false, "required": true, "schema": {"type": "string"}}, "section": {"nullable": true, "required": true, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["DOCUMENT"], "type": "string"}}, "type": "object"}
```

DocumentHit

```
{"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"pattern": "^+cs2-[0-9]{4}$", "type": "string"}}, "content": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "doc_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "model_code": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "score": {"nullable": false, "required": true, "schema": {"maximum": 1, "minimum": -1, "type": "number"}}, "section": {"nullable": true, "required": true, "schema": {"type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}
```

DocumentSearchRequest

```
{"additional_properties": false, "fields": {"model_code": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "query": {"nullable": false, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "top_k": {"nullable": false, "required": false, "schema": {"default": 4, "maximum": 10, "minimum": 1, "type": "integer"}}, "type": "object"}
```

DocumentType

```
{"enum": ["MANUAL", "SPEC", "TROUBLESHOOT"], "type": "string"}
```

ErrorCode

```
{"enum": ["APPROVAL_ALREADY_DECIDED", "DEPENDENCY_NOT_READY", "IDEMPOTENCY_CONFLICT", "INCIDENT_ALREADY_PROCESSED", "INCIDENT_ALREADY_RUNNING", "INTERNAL_ERROR", "LEGACY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MODEL_NOT_READY", "POLICY_REJECTED", "RESOURCE_NOT_FOUND", "UNAUTHORIZED", "VALIDATION_ERROR"], "type": "string"}
```

ErrorResponse

```
{"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "details": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "message": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}
```

EvaluationListResponse

```
{"additional_properties": false, "fields": {"items": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"accuracy": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "correct": {"nullable": false, "required": true, "schema": {"minimum": 0, "type": "integer"}}, "defense_passed": {"nullable": false, "required": true, "schema": {"minimum": 0, "type": "integer"}}, "defense_total": {"nullable": false, "required": true, "schema": {"minimum": 0, "type": "integer"}}, "executed_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "items": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"actual_result": {"nullable": true, "required": false, "schema": {"type": "any"}}, "actual_visualization": {"nullable": true, "required": false, "schema": {"additional_properties": false, "fields": {"chart_type": {"nullable": false, "required": true, "schema": {"enum": ["bar", "histogram", "line", "table"], "type": "string"}}, "x": {"nullable": true, "required": false, "schema": {"type": "string"}}, "y": {"nullable": true, "required": false, "schema": {"type": "string"}}, "type": "object"}}, "attempt_count": {"nullable": false, "required": true, "schema": {"minimum": 0, "type": "integer"}}, "case_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "case_type": {"nullable": false, "required": true, "schema": {"enum": ["DEFENSE", "GOLD"], "type": "string"}}, "expected_result": {"nullable": true, "required": false, "schema": {"type": "any"}}, "expected_visualization": {"nullable": true, "required": false, "schema": {"additional_properties": false, "fields": {"chart_type": {"nullable": false, "required": true, "schema": {"enum": ["bar", "histogram", "line", "table"], "type": "string"}}, "x": {"nullable": true, "required": false, "schema": {"type": "string"}}, "y": {"nullable": true, "required": false, "schema": {"type": "string"}}, "type": "object"}}, "generated_sql": {"nullable": true, "required": false, "schema": {"type": "string"}}, "latency_ms": {"nullable": true, "required": false, "schema": {"minimum": 0, "type": "integer"}}, "passed": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "question": {"nullable": true, "required": false, "schema": {"type": "string"}}, "reason": {"nullable": true, "required": false, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "model": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "prompt_version": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "provider": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "temperature": {"nullable": false, "required": true, "schema": {"type": "number"}}, "total": {"nullable": false, "required": true, "schema": {"minimum": 0, "type": "integer"}}, "type": "object"}, "type": "array"}}, "page": {"nullable": false, "required": true, "schema": {"minimum": 1, "type": "integer"}}
```

```
, "size": {"nullable": false, "required": true, "schema": {"maximum": 100, "minimum": 1, "type": "integer"}}, "total": {"nullable": false, "required": true, "schema": {"minimum": 0, "type": "integer"}}, "type": "object"}
```

EvidenceItem

```
{"discriminator": {"type": "type", "type": "discriminated_union", "variants": {"ALARM": {"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["ALARM"], "type": "string"}}, "type": "object"}, "DOCUMENT": {"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "excerpt": {"nullable": false, "required": true, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["DOCUMENT"], "type": "string"}}, "type": "object"}, "GRAPH": {"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"type": "string"}}, "graph_revision": {"nullable": false, "required": true, "schema": {"pattern": "^0-9a-f{64}$", "type": "string"}}, "relation_id": {"nullable": false, "required": true, "schema": {"pattern": "^REL-[0-9a-f]{20}$", "type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["GRAPH"], "type": "string"}}, "type": "object"}, "METROLOGY": {"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["METROLOGY"], "type": "string"}}, "rules": [{"metrology_alarm_result is forbidden"}, {"type": "object"}], "TRACE": {"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["TRACE"], "type": "string"}}, "type": "object"}}}
```

FaultHypothesis

```
{"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}
```

GraphAskEvidence

```
{"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "graph_revision": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "relation_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}
```

GraphContext

```
{"additional_properties": false, "fields": {"adjacent_process_step_ids": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "area": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "model_code": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "parameter_ids": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "process_step_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "relation_ids": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "type": "object"}
```

GraphEvidence

```
{"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"type": "string"}}, "graph_revision": {"nullable": false, "required": true, "schema": {"pattern": "^0-9a-f{64}$", "type": "string"}}, "relation_id": {"nullable": false, "required": true, "schema": {"pattern": "^REL-[0-9a-f]{20}$", "type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["GRAPH"], "type": "string"}}, "type": "object"}
```

GraphNode

```
{"additional_properties": false, "fields": {"business_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "label": {"nullable": false, "required": true, "schema": {"enum": ["Area", "Chamber", "Equipment", "EquipmentModel", "Parameter", "ProcessStep"], "type": "string"}}, "name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "node_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "properties": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "type": "object"}
```

GraphQueryRequest

```
{"additional_properties": false, "fields": {"question": {"nullable": false, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "type": "object"}
```

GraphQueryResponse

```
{"additional_properties": false, "fields": {"columns": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "error_msg": {"nullable": true, "required": false, "schema": {"type": "string"}, "type": "array"}}, "type": "object"}
```

```
": {"type": "string"}}, "generated_cypher": {"nullable": true, "required": false, "schema": {"type": "string"}}, "is_rejected": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "is_valid": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "latency_ms": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "question": {"nullable": false, "required": true, "schema": {"type": "string"}}, "reject_reason": {"nullable": true, "required": false, "schema": {"type": "string"}}, "row_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "rows": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": true, "fields": {}, "type": "object"}, "type": "array"}}, "type": "object"}
```

GraphRelationship

```
{"additional_properties": false, "fields": {"from_node_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "relation_id": {"nullable": false, "required": true, "schema": {"pattern": "^REL-[0-9a-f]{20}$", "type": "string"}}, "to_node_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}
```

GroupeMetricResult

```
{"additional_properties": false, "fields": {"group": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "value": {"nullable": false, "required": false, "schema": {"type": "union", "variants": [{"type": "integer"}, {"type": "number"}, {"type": "null"}]}}, "type": "object"}
```

HTTPValidationError

```
{"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object"}
```

HealthResponse

```
{"additional_properties": false, "fields": {"status": {"nullable": false, "required": true, "schema": {"enum": ["UP"], "type": "string"}}, "type": "object"}
```

MetricPlan

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MetrologyAskEvidence

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MetrologyEvidence

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NIQueryOutcome

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AUDIT EVENTS
DETECTION_COMPLETED / LOT_HIST / A
AGENT_RUN_STARTED / AGENT_RUN / C
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HYPOTHESIS_GENERATED / AGENT_RUN / C
APPROVAL_REQUESTED / APPROVAL / C
APPROVAL_DECIDED / APPROVAL / C
ACTION_SENT / ACTION / C delivery service
ACTION_SEND_FAILED / ACTION / C delivery service
AGENT_RUN_COMPLETED / AGENT_RUN / C
AGENT_RUN_FAILED / AGENT_RUN / C